

Chapter 53

Population Ecology

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

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Overview

- **Population ecology** is the study of populations in relation to environment, including environmental influences on density and distribution, age structure, and population size.
- A **population** is a group of individuals of a single species living in the same general area.

Dynamic biological processes influence population density, dispersion, and demographics

- **Density** is the number of **individuals per unit area** or volume.
- **Dispersion** is the **pattern of spacing** among individuals within the boundaries of the population.
- Density is the result of an interplay between processes that add individuals to a population and those that remove individuals.

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- **Immigration** is the influx of new individuals from other areas.
 - **Emigration** is the movement of individuals out of a population.
 - In most cases, it is impractical or impossible to count all individuals in a population. **Sampling techniques** can be used to **estimate** densities and **total population** sizes.
 - Population size can be estimated by either extrapolation from small samples, an index of population size, or the **Mark-Recapture Method**.

Population dynamics

Births



**Births and immigration
add individuals to
a population.**



Immigration

Deaths



**Deaths and emigration
remove individuals
from a population.**

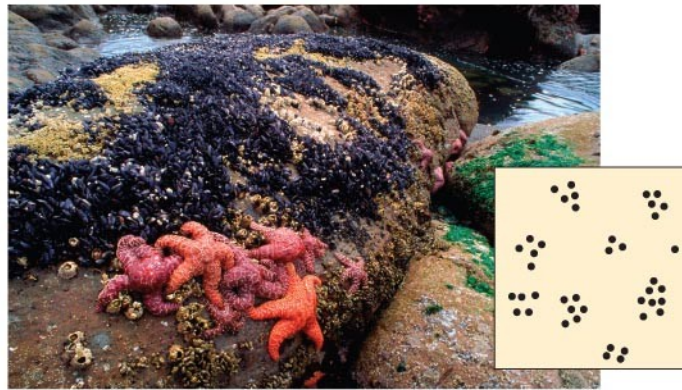


Emigration

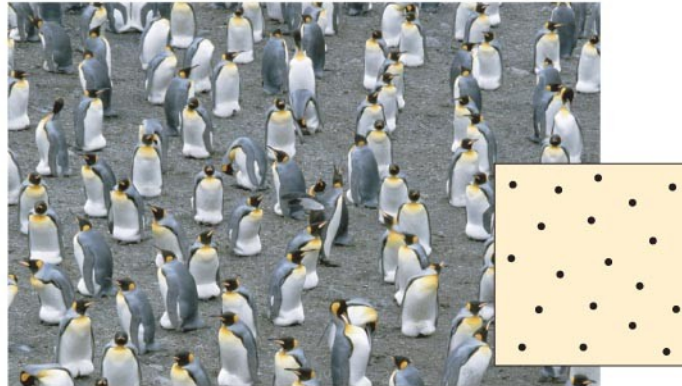
Patterns of Dispersion

- Environmental and social factors influence spacing of individuals in a population.
- In a *clumped* dispersion, individuals aggregate in patches. A clumped dispersion may be influenced by resource availability and behavior.
- A *uniform* dispersion is one in which individuals are evenly distributed. It may be influenced by social interactions such as **territoriality**.
- In a *random* dispersion, the position of each individual is independent of other individuals. It occurs in the absence of strong attractions or repulsions.

Patterns of dispersion within a population's geographic range



(a) **Clumped**



(b) **Uniform**



(c) **Random**

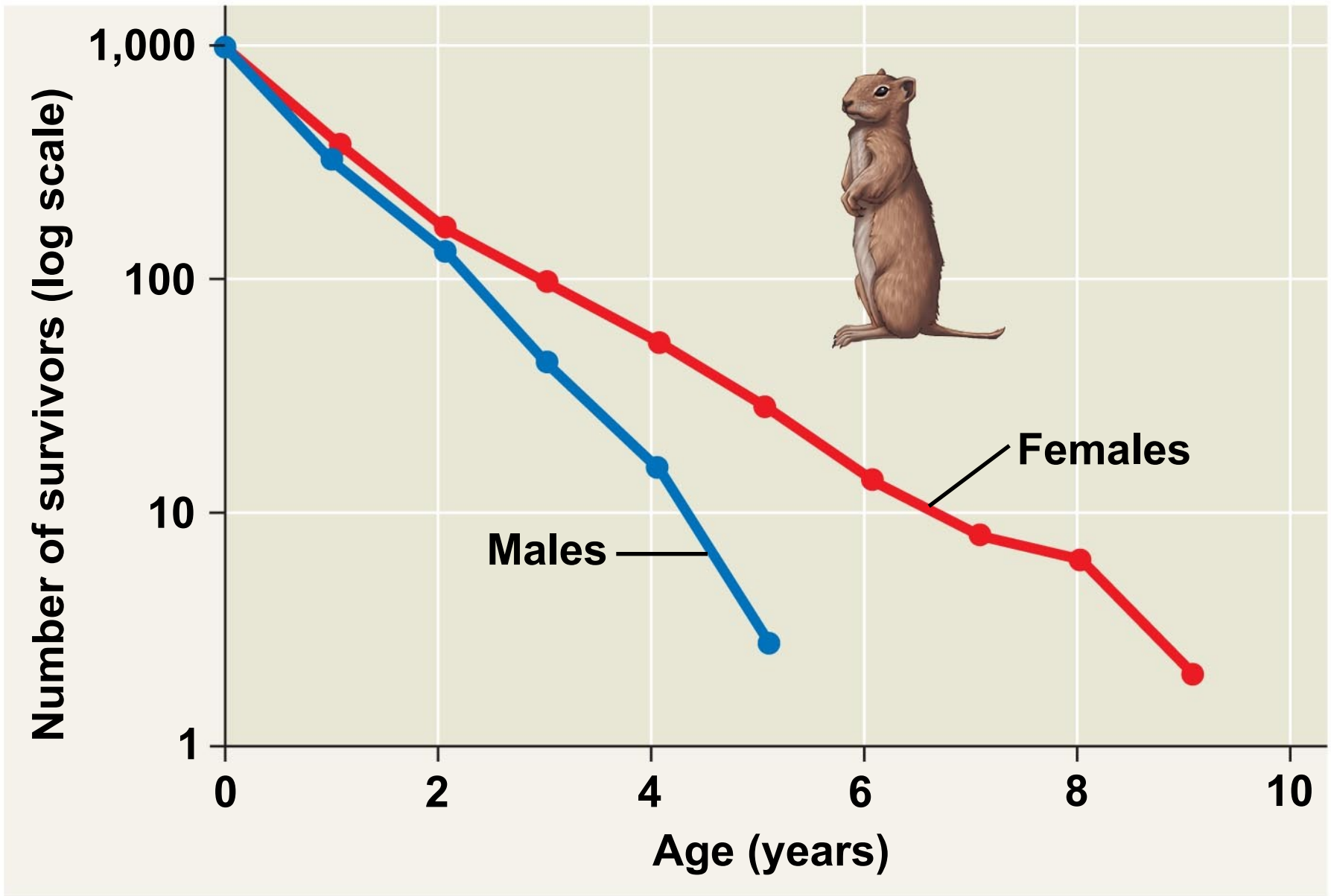
Demographics

- **Demography** is the study of the vital statistics of a population and how they change over time.
- Death rates and birth rates are of particular interest to demographers.

Life Tables & Survivorship Curves

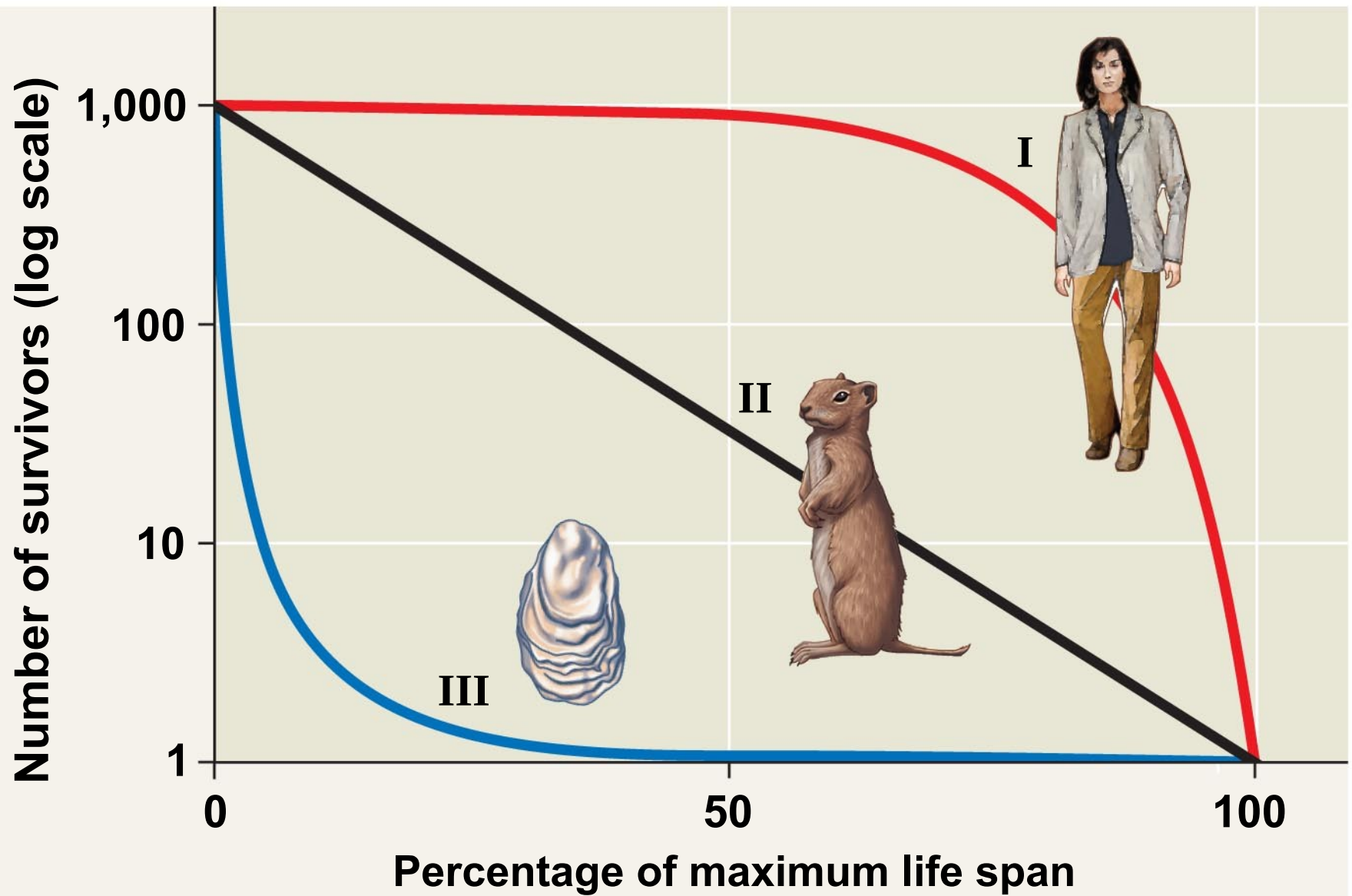
- A **life table** is an age-specific summary of the survival pattern of a population.
- It is best made by following the fate of a **cohort**, a group of individuals of the same age.
- A **survivorship curve** is a graphic way of representing the data in a life table.

Survivorship curves for squirrels shows relatively constant death rate



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- **Survivorship curves** can be classified into three general types:
 - **Type I**: low death rates during early and middle life, then an increase among older age groups
 - **Type II**: the death rate is constant over the organism's life span
 - **Type III**: high death rates for the young, then a slower death rate for survivors

Survivorship Curves



Reproductive Rates

- For species with sexual reproduction, demographers often concentrate on females in a population.
- A **reproductive table**, or **fertility schedule**, is an age-specific summary of the reproductive rates in a population. It **describes reproductive patterns of a population**.

Life history traits are products of natural selection

- An organism's **life history** comprises the traits that affect its schedule of reproduction and survival:
 - The age at which reproduction begins
 - How often the organism reproduces
 - How many offspring are produced during each reproductive cycle
- Life history traits are evolutionary outcomes reflected in the development, physiology, and behavior of an organism.

Evolution and Life History Diversity

- Life histories are very diverse.
- Species that exhibit **semelparity**, or **big-bang reproduction**, reproduce once and die.
- Species that exhibit **iteroparity**, or **repeated reproduction**, produce offspring repeatedly.
- Highly variable or unpredictable environments likely favor big-bang reproduction, while dependable environments may favor repeated reproduction.

“Trade-offs” and Life Histories

- ***Organisms have finite resources, which may lead to trade-offs between survival and reproduction.***
- In animals, parental care of smaller broods may facilitate survival of offspring.
- Some plants, like the dandelion, produce a large number of small seeds, ensuring that at least some of them will grow and eventually reproduce.
- Other types of plants, like the coconut tree, produce a moderate number of large seeds that provide a large store of energy that will help seedlings become established.

Variation in the size of seed crops in plants



(a) Dandelion



(b) Coconut palm

The exponential model describes population growth in an idealized, unlimited environment

- It is useful to study population growth in an idealized situation.
- Idealized situations help us understand the capacity of species to increase and the conditions that may facilitate this growth.

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- **Zero population growth** occurs when the **birth rate equals the death rate**.
 - Most ecologists use differential calculus to express population growth as growth rate at a particular instant in time:

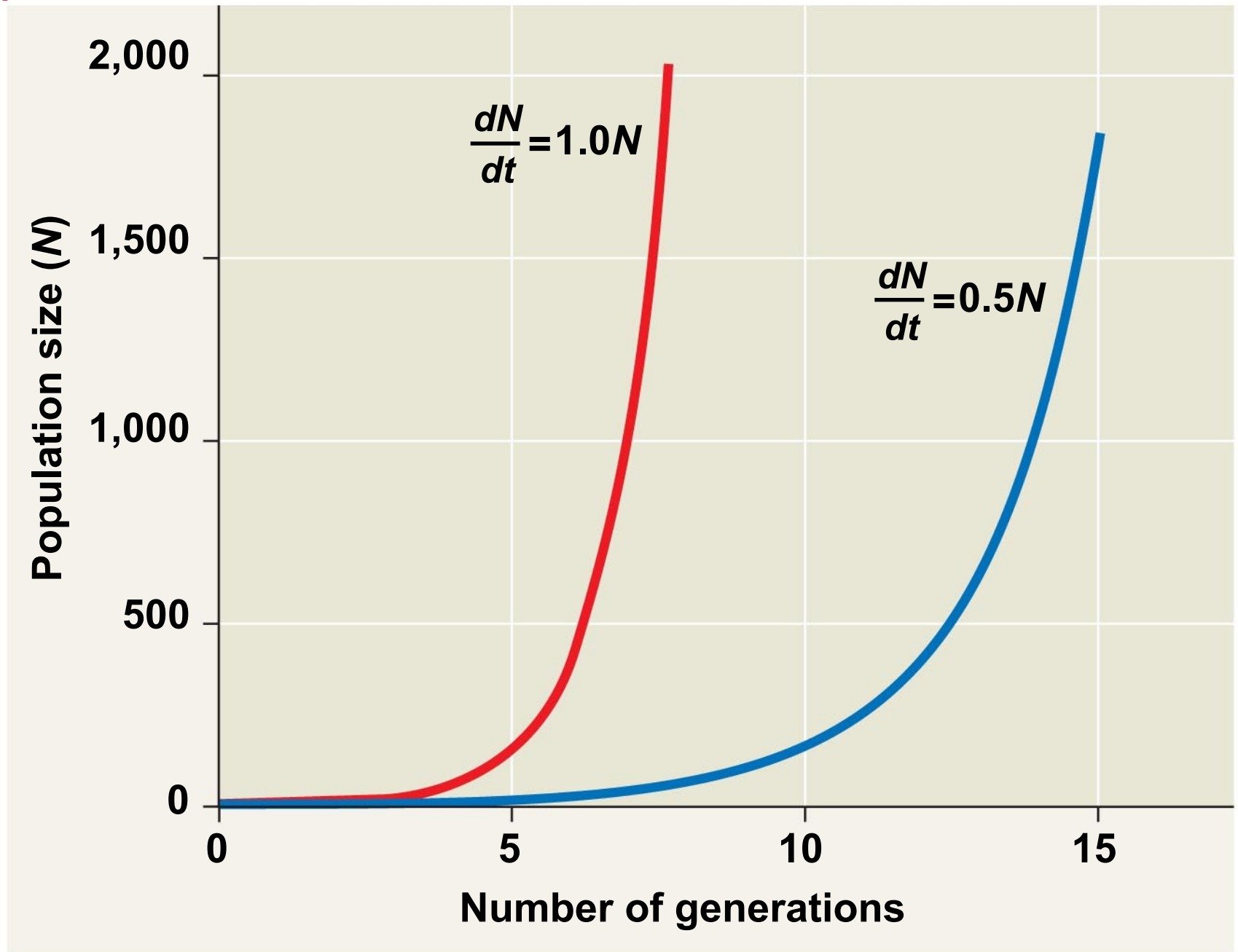
$$\frac{\Delta N}{\Delta t} = rN$$

where N = population size, t = time, and r = per capita rate of increase = birth – death

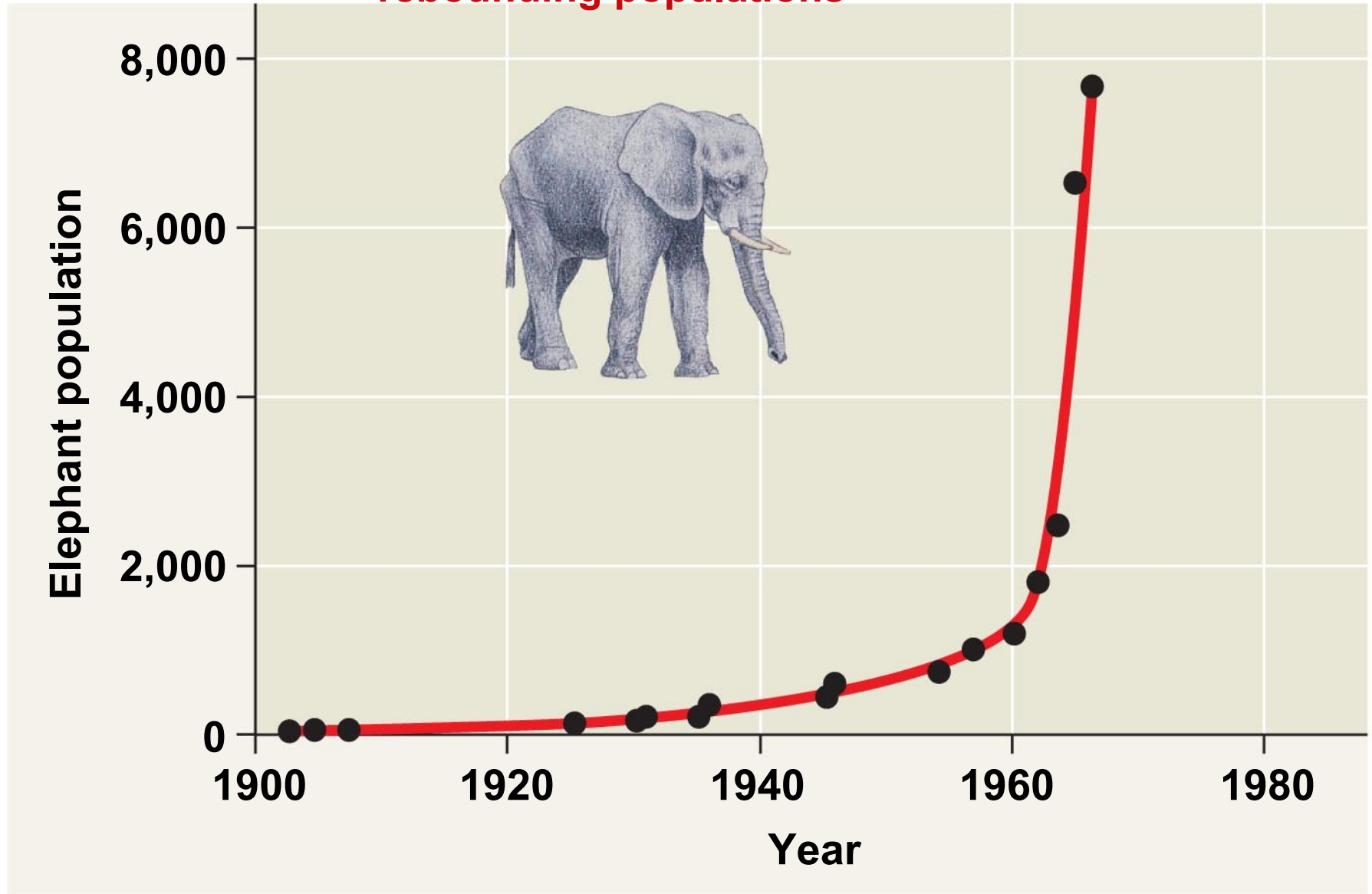
Exponential Growth

- **Exponential population growth** is population increase under idealized conditions.
- Under these conditions, the rate of reproduction is at its maximum, called the intrinsic rate of increase.
- Exponential population growth results in a **J-shaped curve**
- **Exponential Growth is not sustainable.**

Exponential Growth Model



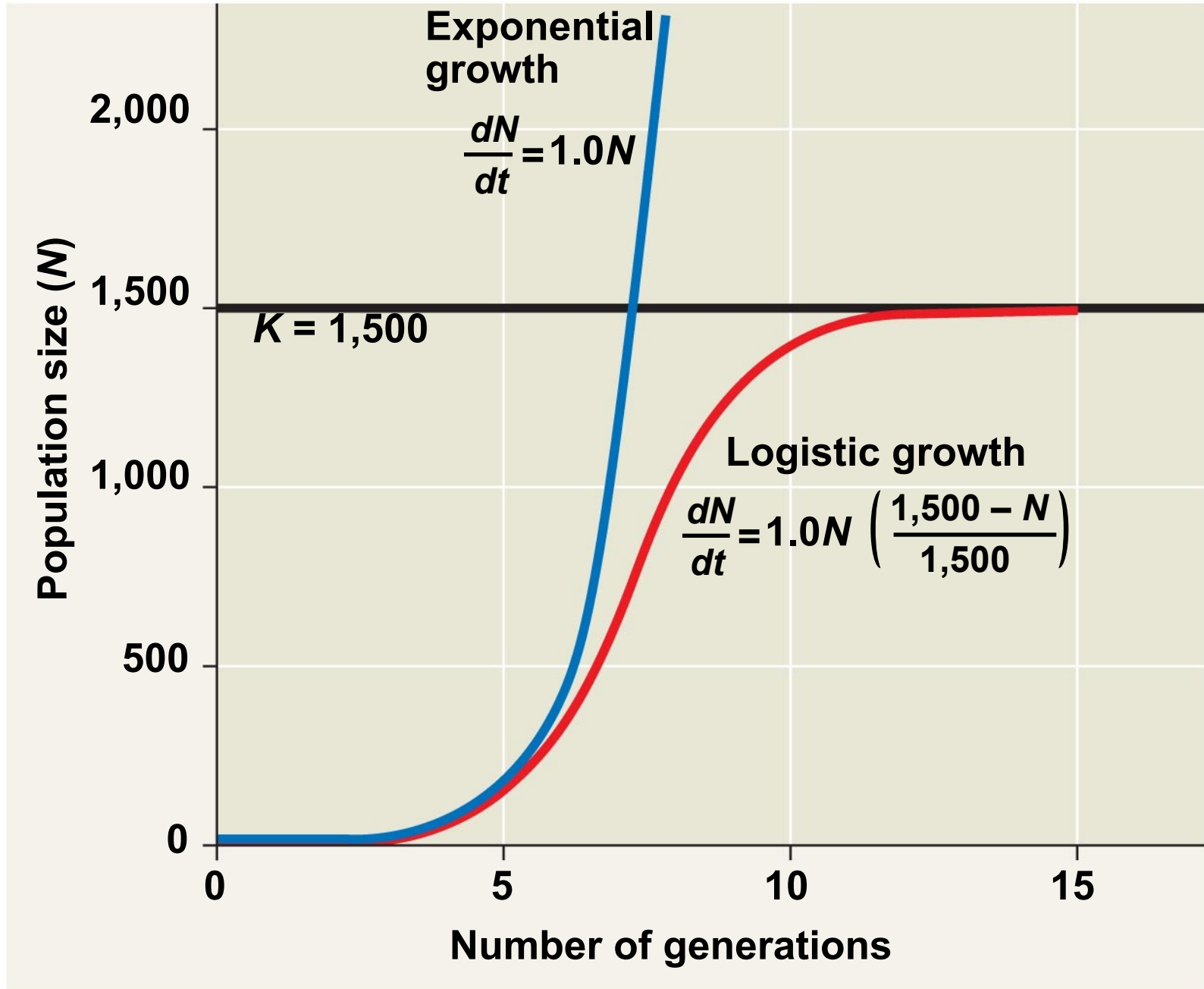
The J-shaped curve of exponential growth characterizes some rebounding populations



The logistic model describes how a population grows more slowly as it nears its carrying capacity

- Exponential growth cannot be sustained for long in any population. A more realistic population model limits growth by incorporating carrying capacity.
- **Carrying capacity** (K) is the maximum population size the environment can support.
- In the **logistic population growth** model, the *rate of increase declines as carrying capacity is reached*.
- The *logistic model* of population growth produces a sigmoid (S-shaped) curve.

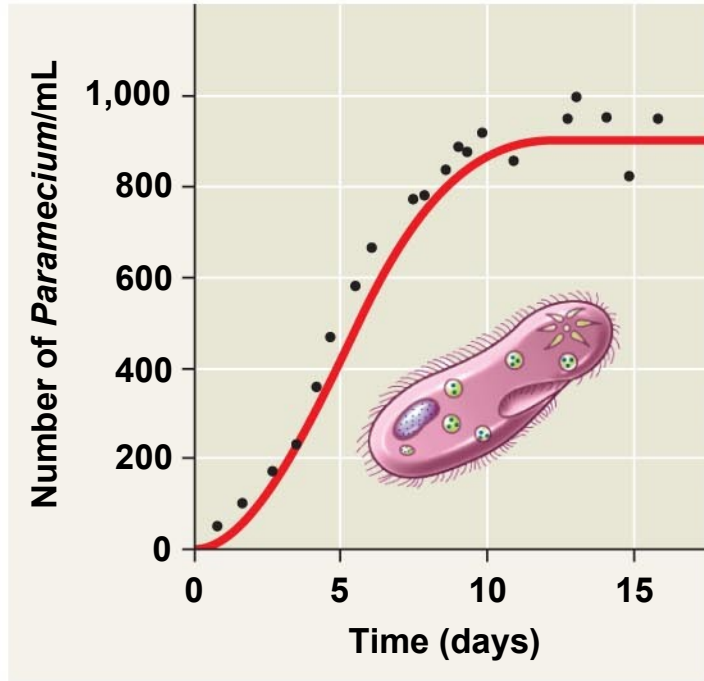
Logistic Growth model



The Logistic Model and Real Populations

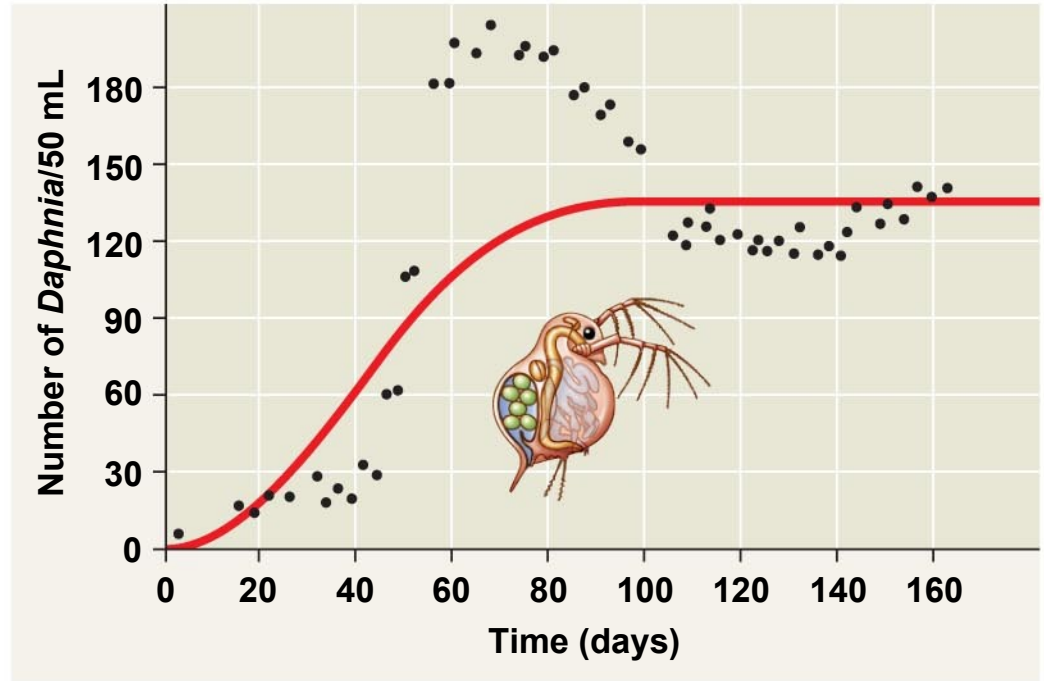
- The growth of laboratory populations of paramecia fits an S-shaped curve.
- These organisms are grown in a constant environment lacking predators and competitors.
- Some populations overshoot K before settling down to a relatively stable density.

The growth of laboratory populations fits an **S-shaped curve** which hovers around the **Carrying Capacity** of the area.



(a) A *Paramecium* population in the lab

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(b) A *Daphnia* population in the lab

The Logistic Model and Life Histories

- Life history traits favored by natural selection may vary with population density and environmental conditions.
- **K-selection** = density-dependent selection, selects for life history traits that are sensitive to population density.
- **r-selection** = or density-independent selection, selects for life history traits that maximize reproduction.

Many factors that regulate population growth are density dependent

- There are two general questions about regulation of population growth:
 - What environmental factors stop a population from growing indefinitely?
 - Why do some populations show radical fluctuations in size over time, while others remain stable?

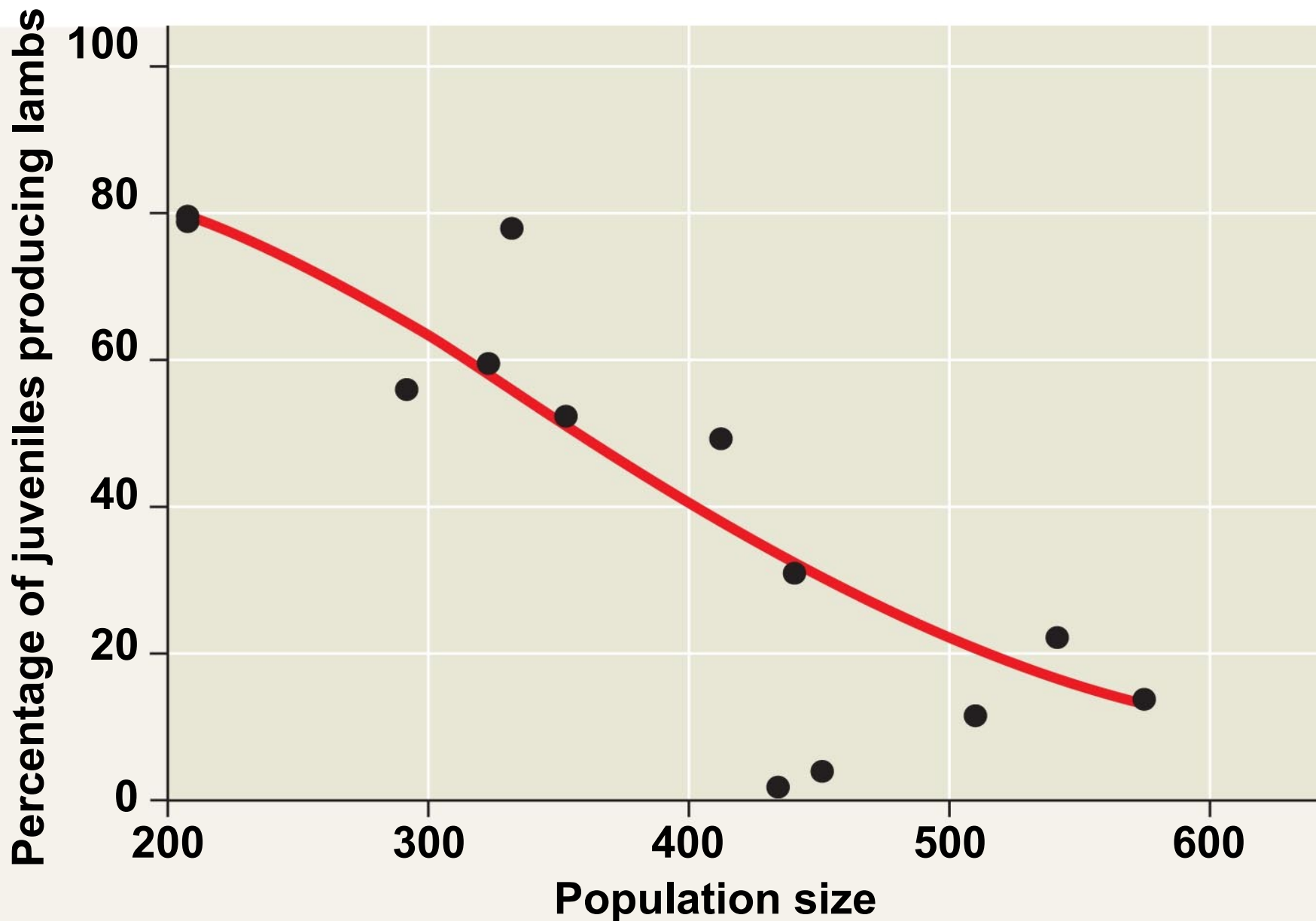
Population Change and Population Density

- In **density-independent** populations, birth rate and death rate do not change with population density.
- In **density-dependent** populations, birth rates fall and death rates rise with population density.

Density-Dependent Population Regulation

- ***Density-dependent*** birth and death rates are an example of *negative feedback that regulates population growth*.
- They are affected by many factors, such as *competition for resources, territoriality, disease, predation, toxic wastes, and intrinsic factors*.
- In crowded populations, increasing population density intensifies competition for resources and results in a lower birth rate.

Decreased reproduction at high population densities



Territoriality

- In many vertebrates and some invertebrates, competition for territory may limit density.
- Cheetahs are highly territorial, using chemical communication to warn other cheetahs of their boundaries.

Territoriality



(a) Cheetah marking its territory



(b) Gannets

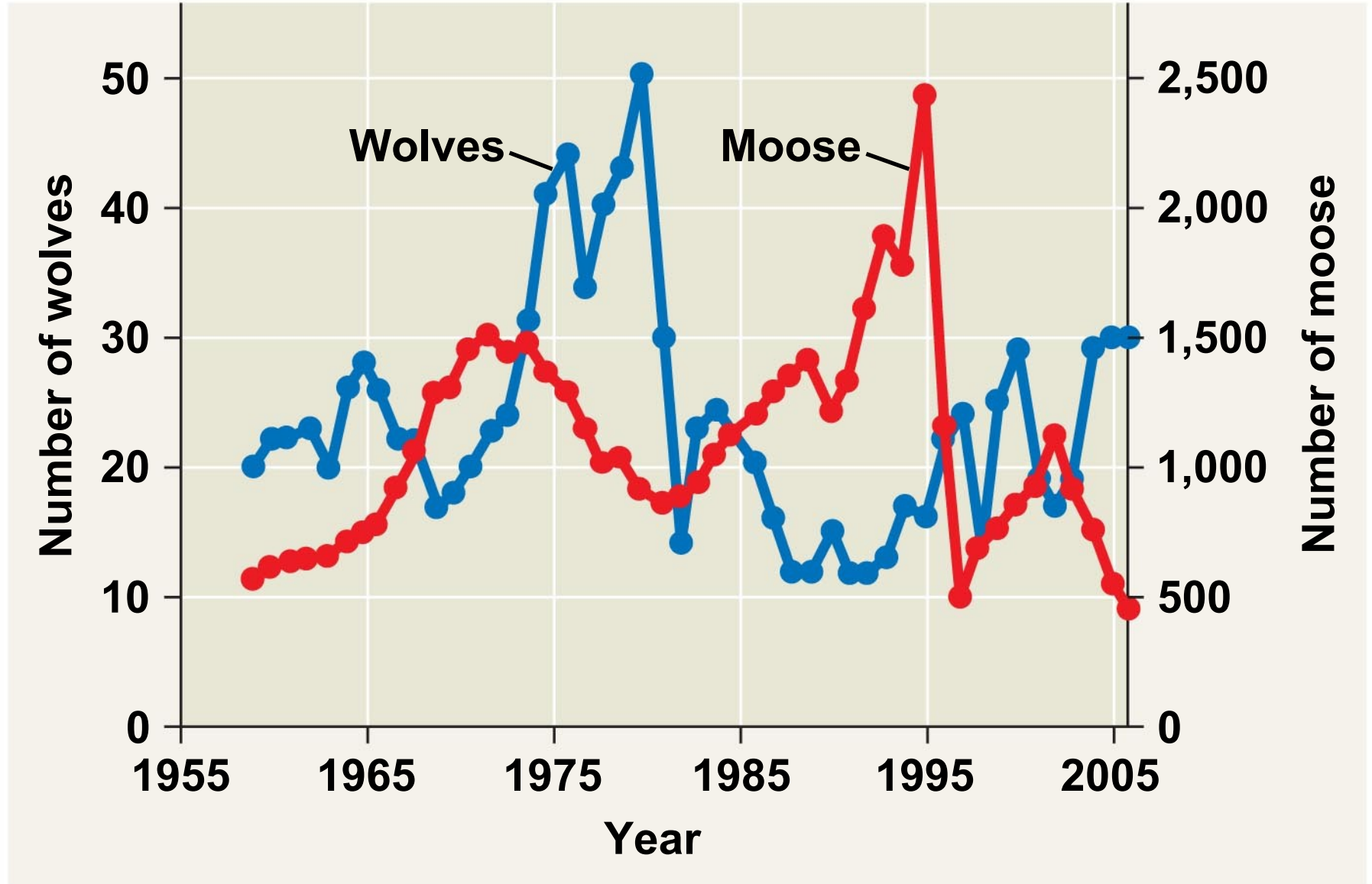
Disease, Predation, & Toxic Wastes

- Population density can influence the health and survival of organisms. In dense populations, pathogens can spread more rapidly.
- As a prey population builds up, predators may feed preferentially on that species.
- Accumulation of toxic wastes can contribute to density-dependent regulation of population size.
- For some populations, **intrinsic** (physiological) **factors** appear to regulate population size.

Population Dynamics

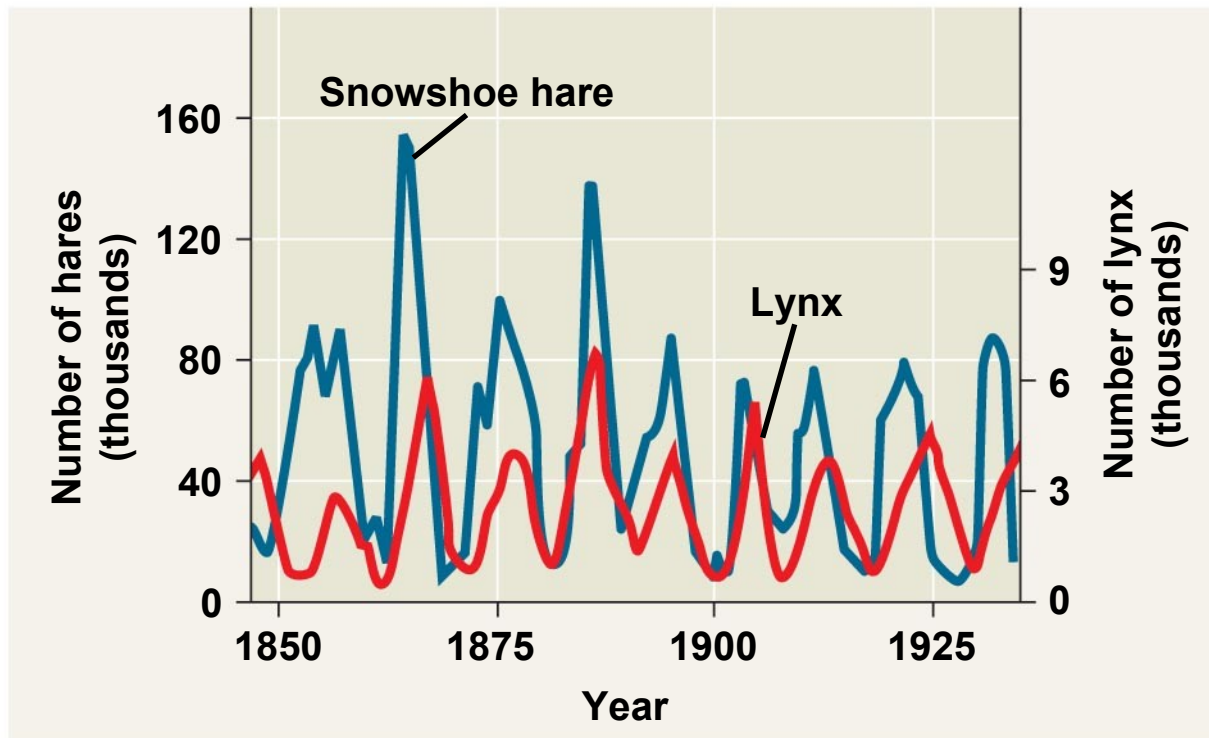
- The study of **population dynamics** focuses on the *complex interactions between biotic and abiotic factors that cause variation in population size*.
- Long-term population studies have challenged the hypothesis that populations of large mammals are relatively stable over time.
- Weather can affect population size over time.

Changes in predation pressure can drive population fluctuations



Population Cycles: Scientific Inquiry

- Some populations undergo regular boom-and-bust cycles.
- Lynx populations follow the 10 year boom-and-bust cycle of hare populations.
- Three hypotheses have been proposed to explain the hare's 10-year interval.



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- **Hypothesis 1:** The hare's population cycle follows a cycle of winter food supply.
 - If this hypothesis is correct, then the cycles should stop if the food supply is increased.
 - Additional food was provided experimentally to a hare population, and the whole population increased in size but continued to cycle.
 - No hares appeared to have died of starvation.

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- **Hypothesis 2:** The hare's population cycle is driven by pressure from other predators.
 - In a study conducted by field ecologists, 90% of the hares were killed by predators.
 - These data support this second hypothesis.

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- **Hypothesis 3:** The hare's population cycle is linked to sunspot cycles.
 - Sunspot activity affects light quality, which in turn affects the quality of the hares' food.
 - There is good correlation between sunspot activity and hare population size.
 - The results of all these experiments suggest that both predation and sunspot activity regulate hare numbers and that food availability plays a less important role.

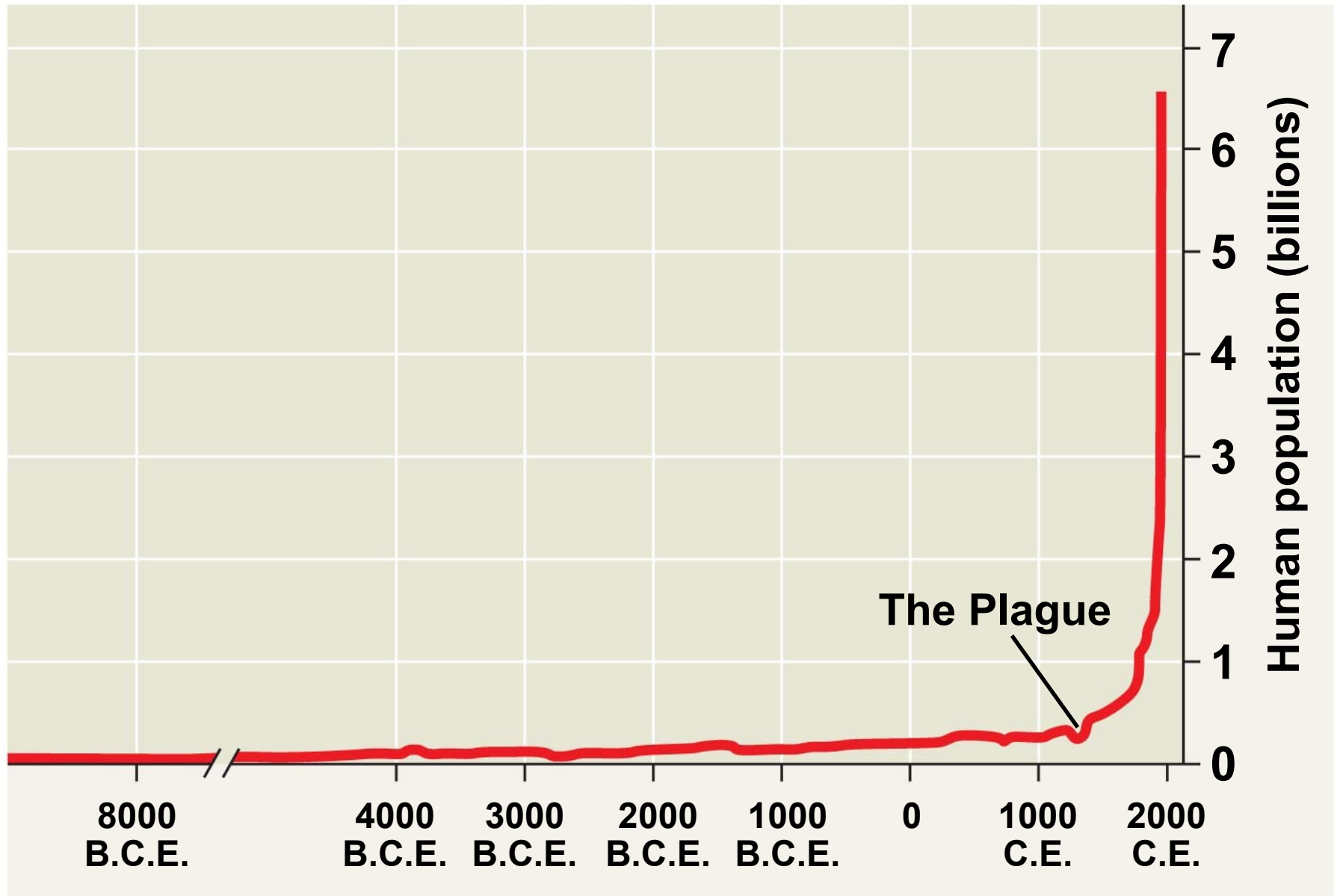
Immigration, Emigration, and Metapopulations

- **Metapopulations** are groups of populations linked by immigration and emigration.
- High levels of immigration combined with higher survival can result in greater stability in populations.

The human population is no longer growing exponentially but is still increasing rapidly

- No population can grow indefinitely, and humans are no exception.
- The human population increased relatively slowly until about 1650 and then began to grow exponentially.
- Though the global population is still growing, the rate of growth began to slow during the 1960s.
- Most of the current global population growth is concentrated in developing countries.

Human population growth



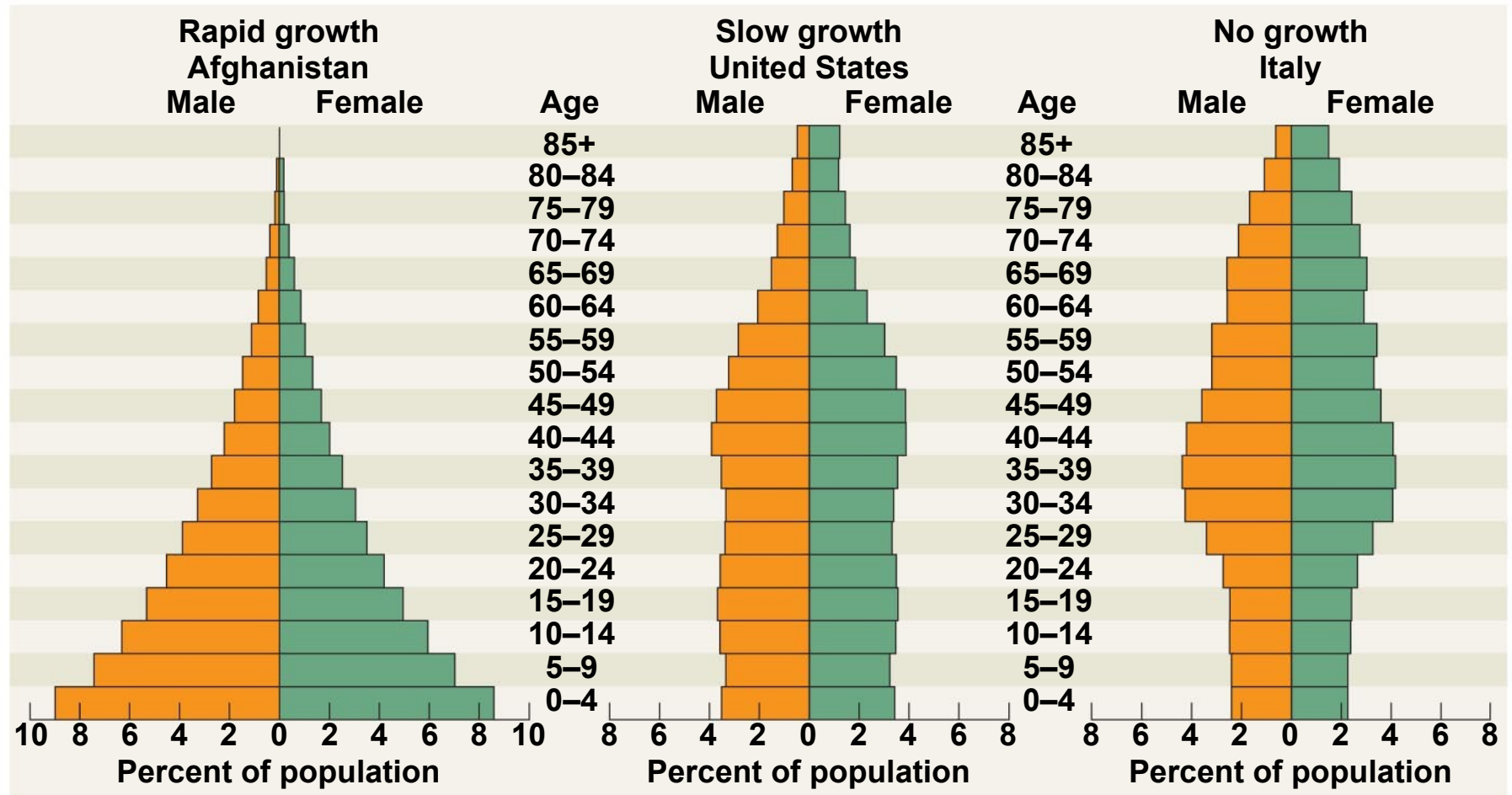
Regional Patterns of Population Change

- To maintain population stability, a regional human population can exist in one of two configurations:
 - Zero population growth =
High birth rate – High death rate
 - Zero population growth =
Low birth rate – Low death rate
- The **demographic transition** is the move from the first state toward the second state.

Age Structure

- One important demographic factor in present and future growth trends is a country's **age structure**.
- Age structure is the relative number of individuals at each age.
- Age structure diagrams can predict a population's growth trends.
- They can illuminate social conditions and help us plan for the future.

Age-structure pyramids for the human population of three countries



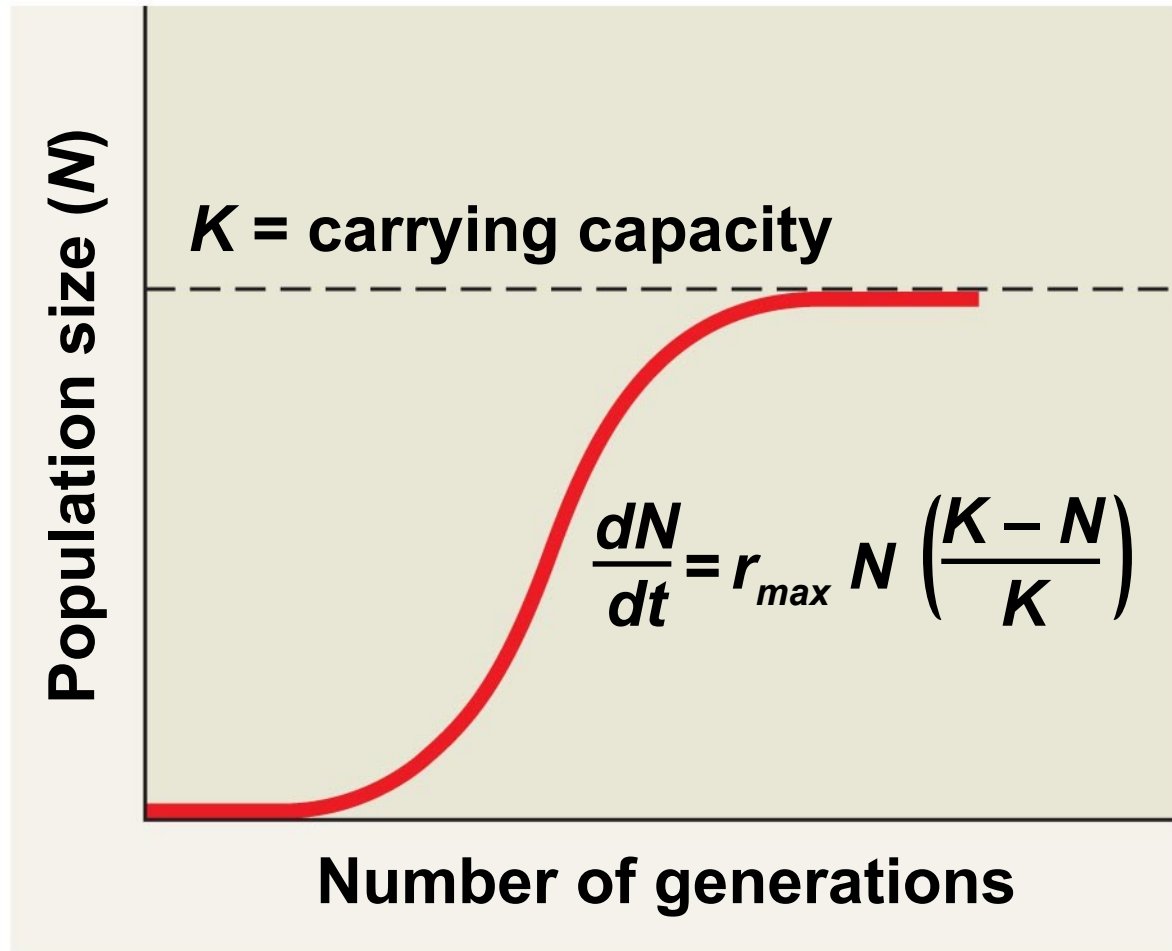
Estimates of Earth's Carrying Capacity

- How many humans can the biosphere support?
- The carrying capacity of Earth for humans is uncertain.
- The average estimate is 10–15 billion.

Limits on Human Population Size

- The **ecological footprint** concept summarizes the aggregate land and water area needed to sustain the people of a nation.
- It is one measure of how close we are to the carrying capacity of Earth.
- Countries vary greatly in footprint size and available ecological capacity.
- *Our carrying capacity could potentially be limited by food, space, nonrenewable resources, or buildup of wastes.*

Review: Population Growth Curve



You should now be able to:

1. Define and distinguish between the following sets of terms: density and dispersion; clumped dispersion, uniform dispersion, and random dispersion; life table and reproductive table; Type I, Type II, and Type III survivorship curves; semelparity and iteroparity; *r*-selected populations and *K*-selected populations.
2. Explain how ecologists may estimate the density of a species.

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3. Explain how limited resources and trade-offs may affect life histories.
 4. Compare the exponential and logistic models of population growth.
 5. Explain how density-dependent and density-independent factors may affect population growth.
 6. Explain how biotic and abiotic factors may work together to control a population's growth.